Instructions

- Submit the assignments in time via online form here.
- The solutions MUST be done in LaTeXand handed as pdf. No handwritten assignments will be accepted.
- For python based problems, provide the codes as well as any generated figure. Seek your instructor's or peers' help if you need it. Responses during the viva that amount to: '...that's what the AI gave me', would result in a 0.
- Equations must be numbered and properly formatted.
- Expect some simple questions from the readings/viewings in the exam as well.
- Your final marks in the assignment will be contingent on the viva.

Project I — Multipole Expansions

1. In this assignment, you will investigate the electrostatic interaction between two neutral charge distributions. You will calculate the exact Coulomb interaction energy and compare it with the dipole–dipole approximation obtained from a multipole expansion.

   The aim is to understand when a localized charge distribution can be approximated by an electric dipole and how the interaction depends on separation and orientation.

   Exact Coulomb Interaction

   Consider two electric dipoles. Each dipole consists of charges $+q$ and $-q$ separated by a distance $2a$.

   Let the first dipole be centered at the origin, with its charges located at

   $$\mathbf{r}_{1+} = (-a, 0, 0), \qquad \mathbf{r}_{1-} = (a, 0, 0).$$

   The second dipole is centered at $(R, 0, 0)$, with charges at

   $$\mathbf{r}_{2+} = (R - a, 0, 0), \qquad \mathbf{r}_{2-} = (R + a, 0, 0).$$

   Assume $R > 2a$.

   (a) Write a Python function that calculates the electrostatic interaction energy between the two dipoles by summing the Coulomb interaction between all pairs of charges belonging to different dipoles.

   (b) Calculate the interaction energy for several values of R, keeping q and a fixed.

   (c) Plot the interaction energy as a function of R. Describe whether the interaction is attractive or repulsive for this configuration.

   The exact interaction energy is

   $$U_{\text{exact}} = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^{2} \sum_{j=1}^{2} \frac{q_i q_j}{|\mathbf{r}_{1i} - \mathbf{r}_{2j}|}, \tag{1}$$

   where the sums run over the charges in the two dipoles.

   The Dipole–Dipole Approximation

   When the separation between localized charge distributions is much larger than their dimensions, the leading interaction can be expressed in terms of their electric dipole moments.

   The dipole moment is defined as

   $$\mathbf{p} = \sum_i q_i \mathbf{r}_i. \tag{2}$$

   For two dipoles separated by $\mathbf{R} = R\hat{\mathbf{R}}$, the dipole–dipole interaction energy is

   $$U_{\text{dd}} = \frac{1}{4\pi\epsilon_0 R^3} \left[\mathbf{p}_1 \cdot \mathbf{p}_2 - 3(\mathbf{p}_1 \cdot \hat{\mathbf{R}})(\mathbf{p}_2 \cdot \hat{\mathbf{R}}) \right]. \tag{3}$$

   (a) Calculate the dipole moment of each charge distribution.

   (b) Implement the dipole–dipole interaction energy in Python for the configuration in the previous section.

   (c) Plot the exact interaction energy and the dipole–dipole approximation on the same graph.

   (d) Explain why the two results approach one another as the separation increases.

Distance Dependence and Approximation Error

Investigate how the accuracy of the dipole approximation depends on the separation between the two dipoles.

(a) Calculate the relative error

$$\delta(R) = \frac{|U_{\text{exact}} - U_{\text{dd}}|}{|U_{\text{exact}}|}. \tag{4}$$

(b) Plot the relative error against R using logarithmic axes.

(c) Estimate how the error depends on R at large separations.

(d) Explain why the dipole approximation becomes less reliable when the separation is comparable to the size of the charge distributions.

Orientation Dependence

Now rotate the second dipole while keeping its center at $(R, 0, 0)$. Let the first dipole moment point along the positive x-axis and let the second dipole moment make an angle θ with that axis.

The dipole moments are

$$\mathbf{p}_1 = p\hat{\mathbf{x}}, \qquad \mathbf{p}_2 = p(\cos\theta\,\hat{\mathbf{x}} + \sin\theta\,\hat{\mathbf{y}}).$$

(a) Modify your Python code to calculate the exact interaction energy for different orientations of the second dipole.

(b) Calculate the dipole–dipole interaction energy for the same orientations.

(c) Plot the interaction energy as a function of θ over the interval $0 \leq \theta \leq 2\pi$.

(d) Identify orientations for which the interaction is attractive, repulsive, or zero in the dipole approximation.

(e) Compare the exact and approximate results and discuss any differences.